

```
1 package LopVaDoiTuong_2;
2
3 public class SinhVien {
4     private String ma, hoTen;
5     private boolean gioiTinh;
6     private double java, css, html;
7
8     public String toString() {
9         return ma + ", " + hoTen + ", " + (gioiTinh ? "Nam" : "Nu") + ", " + java + ", " + css + ", " + html + ", DTB: " + tinhDTB() + ", " + xepLoai();
10    }
11
12    public double tinhDTB() {
13        return (java + 2 * css + html) / 4;
14    }
15
16    public String xepLoai() {
17        double dtb = tinhDTB();
18        if (dtb < 3.5) return "Kém";
19        if (dtb < 5.0) return "Yếu";
20        if (dtb < 6.5) return "Trung bình";
21        if (dtb < 8.0) return "Khá";
22        if (dtb < 9.0) return "Giỏi";
23        else return "Xuất sắc";
24    }
25
26    public SinhVien(String ma, String hoTen, boolean gioiTinh, double java, double css, double html) {
27        this.ma = ma;
28        this.hoTen = hoTen;
29        this.gioiTinh = gioiTinh;
30        this.java = java;
31        this.css = css;
32        this.html = html;
33    }
34
35    public String getMa() {
36        return ma;
37    }
38
39    public void setMa(String ma) {
40        this.ma = ma;
41    }
42}
```

```
43 public String getHoTen() {
44     return hoTen;
45 }
46
47 public void setHoTen(String hoTen) {
48     this.hoTen = hoTen;
49 }
50
51 public boolean isGioiTinh() {
52     return gioiTinh;
53 }
54
55 public void setGioiTinh(boolean gioiTinh) {
56     this.gioiTinh = gioiTinh;
57 }
58
59 public double getJava() {
60     return java;
61 }
62
63 public void setJava(double java) {
64     this.java = java;
65 }
66
67 public double getCss() {
68     return css;
69 }
70
71 public void setCss(double css) {
72     this.css = css;
73 }
74
75 public double getHtml() {
76     return html;
77 }
78
79 public void setHtml(double html) {
80     this.html = html;
81 }
82 }
```

```
1 package LopVaDoiTuong_2;
2
3 import java.io.*;
4 import java.util.*;
5
6 public class mangSinhVien {
7     private ArrayList<SinhVien> s = new ArrayList<>();
8     public void taoDS_doiFile(String tenFile){
9         try {
10             File f = new File(tenFile);
11             if(f.exists()){
12                 System.out.println("File da duoc mo");
13                 Scanner read = new Scanner(f);
14                 while(read.hasNext()){
15                     String line = read.nextLine();
16                     String s[] = line.split(",");
17                     boolean gt = Boolean.parseBoolean(s[2].trim());
18                     double java = Double.parseDouble(s[3].trim());
19                     double css = Double.parseDouble(s[4].trim());
20                     double html = Double.parseDouble(s[5].trim());
21                     s.add(new SinhVien(s[0],s[1],gt,java,css,html));
22                 }
23             }else System.out.println("File khong co");
24         } catch (Exception e) {
25         }
26     }
27     public void xuatDS(String tieude){
28         System.out.println(tieude);
29         for(SinhVien x:s){
30             System.out.println("\t"+x);
31         }
32     }
33     public double DTS_theoGt(boolean gt){
34         double tong=0,dem=0;
35         for(SinhVien i:s){
36             if(i.isGioiTinh()==gt){
37                 tong+=i.tinhDTS();
38                 dem++;
39             }
40         }
41         return tong/dem;
42     }
```

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39         dem++;
40     }
41     return tong/dem;
42 }
43 public double DTS_gvKha(){
44     double tong=0;
45     int dem=0;
46     for(SinhVien i:s){
47         double sv_kha = i.tinhDTS();
48         if(sv_kha<8 && sv_kha>=6.5){
49             tong+=sv_kha;
50             dem++;
51         }
52     }
53     if(dem == 0) return 0;
54     return tong/dem;
55 }
56 public void sapXep(String tieude){
57     for(int i=0;i<s.size()-1;i++){
58         for(int j=i+1;j<s.size();j++){
59             if(s.get(i).tinhDTS()>s.get(j).tinhDTS()){
60                 SinhVien temp = s.get(i);
61                 s.set(i,s.get(j));
62                 s.set(j,temp);
63             }
64         }
65     }
66     xuatDS(tieude);
67 }
68 public int sinhVienGoi(){
69     int dem=0;
70     for(SinhVien i:s){
71         if(i.tinhDTS()>=8 && i.tinhDTS()<9) dem++;
72     }
73     return dem;
74 }
75 }
76
77 public void xepHang(String tieude) {
78     System.out.println(tieude);
79     for(int i=0;i<s.size();i++){
80         int dem=0;
```

